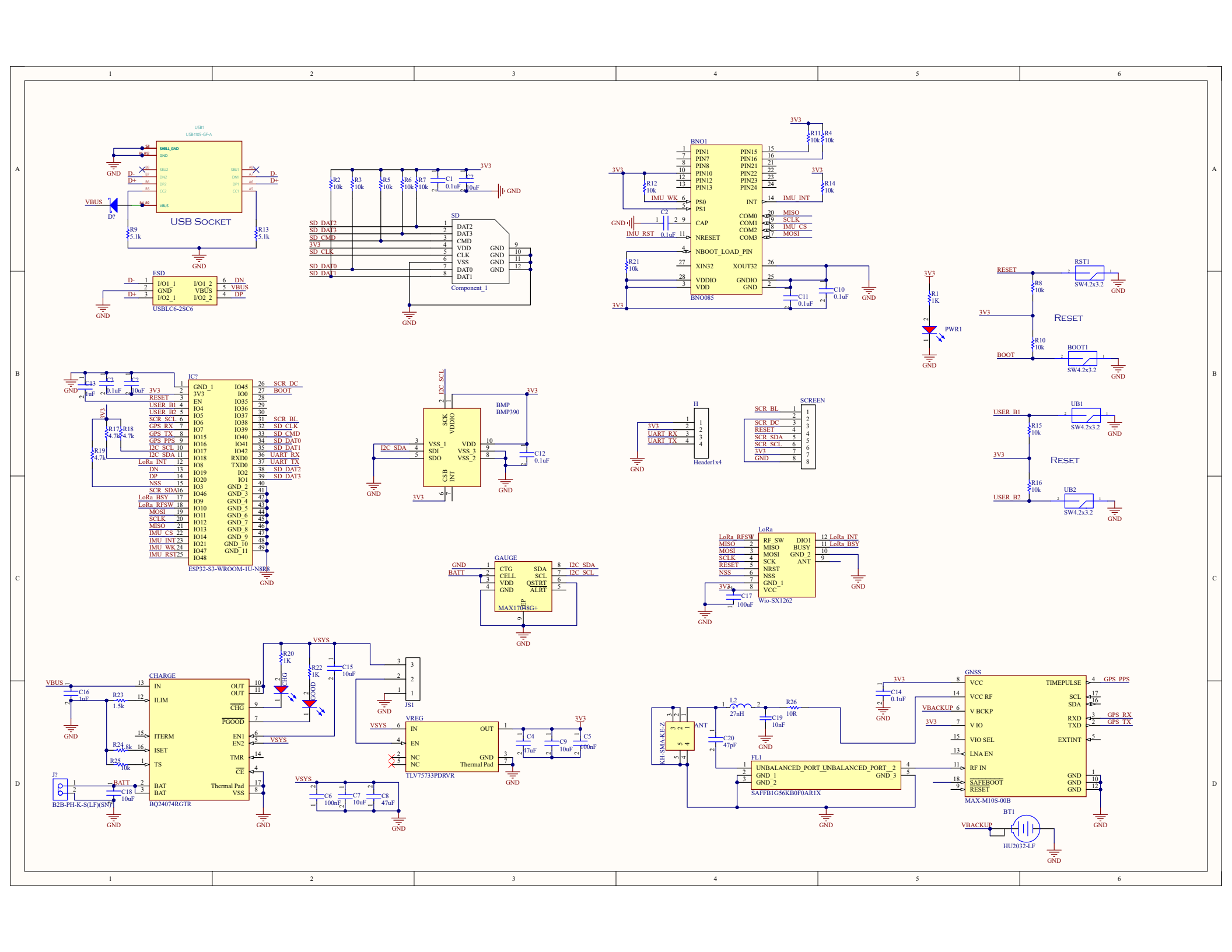

USB1
USB4105-GF-A
SHELL_GND
GND
D-
D+
VBUS
D?
USB Socket
R9
5.1k
R13
5.1k
ESD
I/O1_1
GND
I/O2_1
I/O1_2
VBUS
I/O2_2
DN
DP
USBLC6-2SC6
R2
10k
R3
10k
R5
10k
R6
10k
R7
10k
C1
0.1uF
C2
10uF
3V3
SD_DAT2
SD_DAT3
SD_CMD
SD_CLK
SD_DAT0
SD_DAT1
SD
DAT2
DAT3
CMD
VDD
CLK
VSS
DAT0
DAT1
Component_1
BNO1
PIN1
PIN7
PIN8
PIN10
PIN12
PIN13
PIN15
PIN16
PIN21
PIN22
PIN23
PIN24
INT
PS0
PS1
CAP
NRESET
NBOOT_LOAD_PIN
XIN32
XOUT32
VDDIO
VDD
GNDIO
GND
COM0
COM1
COM2
COM3
MISO
SCLK
IMU_CS
MOSI
IMU_INT
IMU_WK
IMU_RST
R11
10k
R4
10k
R12
10k
R14
10k
R21
10k
C10
0.1uF
C11
0.1uF
BNO085
R1
1K
PWR1
RESET
RST1
SW4.2x3.2
R8
10k
R10
10k
BOOT
BOOT1
IC?
GND_1
3V3
EN
IO4
IO5
IO6
IO7
IO15
IO16
IO17
IO18
IO8
IO19
IO20
IO3
IO46
IO9
IO10
IO11
IO12
IO13
IO14
IO21
IO47
IO48
IO45
IO0
IO35
IO36
IO37
IO38
IO39
IO40
IO41
IO42
RXD0
TXD0
IO2
IO1
GND_2
GND_3
GND_4
GND_5
GND_6
GND_7
GND_8
GND_9
GND_10
GND_11
SCR_DC
BOOT
SD_CLK
SD_CMD
SD_DAT0
SD_DAT1
UART_RX
UART_TX
SD_DAT2
SD_DAT3
USER_B1
USER_B2
SCR_SCL
GPS_RX
GPS_TX
GPS_PPS
I2C_SCL
I2C_SDA
LoRa_INT
DN
DP
NSS
SCR_SDA
LoRa_BSY
LoRa_RFSW
MOSI
SCLK
MISO
IMU_CS
IMU_INT
IMU_WK
IMU_RST
ESP32-S3-WROOM-1U-N8R8
BMP
BMP390
SCK
SDI
SDO
VDDIO
VSS_1
VSS_2
CSB
INT
VDD
C12
0.1uF
H
Header1x4
SCREEN
SCR_BL
SCR_DC
RESET
SCR_SDA
SCR_SCL
UB1
UB2
R15
10k
R16
10k
GAUGE
CTG
CELL
VDD
GND
SDA
SCL
QSTRT
ALRT
MAX17048G+
LoRa
RF_SW
MISO
MOSI
SCK
NRST
NSS
GND_1
VCC
DIO1
BUSY
GND_2
ANT
Wio-SX1262
C17
100uF
CHARGE
IN
ILIM
OUT
CHG
PGOOD
ITERM
ISET
TS
TMR
CE
EN1
EN2
BAT
Thermal Pad
VSS
BQ24074RGTR
R20
1K
R22
1K
C15
10uF
C16
1uF
R23
1.5k
R24
8k
R25
10k
C18
10uF
J?
B2B-PH-K-S(LF)(SN)
JS1
VREG
TLV75733PDRVR
C4
47uF
C9
10uF
C5
100nF
C6
100nF
C7
10uF
C8
47uF
KH-SMAKE-Z
ANT
L2
27nH
R26
10R
C19
10nF
C20
47pF
FL1
UNBALANCED_PORT_UNBALANCED_PORT_2
GND_1
GND_2
GND_3
SAFFB1G56KB0F0AR1X
GNSS
VCC
VCC RF
V BCKP
V IO
VIO SEL
LNA EN
RF IN
SAFEBOOT
RESET
TIMEPULSE
SCL
SDA
RXD
TXD
EXTINT
MAX-M10S-00B
C14
0.1uF
VBACKUP
BT1
HU2032-LF
GPS_PPS
GPS_RX
GPS_TX